

The diagram shows the ESP32 module with the following components and connections:

- Power Supply:** A 3V3 supply is connected to the EN pin (pin 2) and the RST pin (pin 3). A 10K resistor (R1) is connected between the 3V3 supply and the RST pin. A 10uF 16V capacitor (C11) is connected between the 3V3 supply and ground.
- Reset:** A push button (SW1) is connected between the RST pin (pin 3) and ground.
- Capacitors:** A 22uF 10V capacitor (C3) is connected between the 3V3 supply and ground. A 100nF capacitor (C4) is connected between the 3V3 supply and ground.
- Pin Connections:**
 - IO0 (pin 25) is connected to IO1/TXD0 (pin 35).
 - IO1 (pin 35) is connected to IO2 (pin 24).
 - IO2 (pin 24) is connected to IO3/RXD0 (pin 34).
 - IO3 (pin 34) is connected to IO4 (pin 26).
 - IO4 (pin 26) is connected to IO5 (pin 29).
 - IO5 (pin 29) is connected to IO12 (pin 14).
 - IO12 (pin 14) is connected to IO13 (pin 16).
 - IO13 (pin 16) is connected to IO14 (pin 13).
 - IO14 (pin 13) is connected to IO15 (pin 23).
 - IO15 (pin 23) is connected to IO16 (pin 27).
 - IO16 (pin 27) is connected to IO17 (pin 28).
 - IO17 (pin 28) is connected to IO18 (pin 30).
 - IO18 (pin 30) is connected to IO19 (pin 31).
 - IO19 (pin 31) is connected to IO21 (pin 33).
 - IO21 (pin 33) is connected to IO22 (pin 36).
 - IO22 (pin 36) is connected to IO23 (pin 37).
 - IO23 (pin 37) is connected to IO25 (pin 10).
 - IO25 (pin 10) is connected to IO26 (pin 11).
 - IO26 (pin 11) is connected to IO27 (pin 12).
 - IO27 (pin 12) is connected to IO32 (pin 8).
 - IO32 (pin 8) is connected to IO33 (pin 9).
 - IO33 (pin 9) is connected to IO34 (pin 6).
 - IO34 (pin 6) is connected to IO35 (pin 7).
 - IO35 (pin 7) is connected to IO36 (pin 4).
 - IO36 (pin 4) is connected to IO39 (pin 5).
 - IO39 (pin 5) is connected to IO36/SENSOR_VP (pin 15).
 - IO36/SENSOR_VP (pin 15) is connected to IO39/SENSOR_VN (pin 1).

Pins Breakout

2.54 mm (0.1") breakout

J4

CONN-HDR-1x17

1 RST

2 IO36

3 IO39

4 IO34

5 IO35

6 IO32

7 IO33

8 IO25

9 IO26

10 IO27

11

12

13

14

15

16

17

3V3

VBUS

VUSB

VIN

VBAT

J3

1023 1

1022 2

1021 3

1019 4

1018 5

1017 6

1016 7

105 8

104 9

100 10

103 11

101 12

102 13

1015 14

1013 15

1012 16

1014 17

CONN-HDR-1x17

2.54 mm (0.1") breakout

Power Management

The circuit diagram illustrates the power management system. It starts with a solar panel connected to a PMEG3020CPA multiplexer (D1) which can switch between VUSB and VIN. The output goes through a 10µF capacitor (C1) to VEXT. A red LED (D2) indicates charging status. The battery (J1) is connected via a JST PH 2.0mm connector. The CN3082 battery charger (U1) controls the charging process, with feedback from the battery voltage (VBAT) through resistors R10, R11, R12, and R13. A DMP2165UW-7 MOSFET (Q1) drives the buck-boost converter (TPS63900DSKR, U3). The converter takes input from VBUS and provides a regulated 3V3 output. Various capacitors (C2, C5, C6) are used for filtering and stability.

Battery Type	R11	R13	R10	R12	Vcharge	Imin
Li-Ion/Pol	100K	10K	365K	510K	4.2V	0mA
NIMHx3	200K	10K	365K	510K	4.2V	19mA
NIMHx4	200K	10K	510K	390K	5.6V	19mA
LiFePO4	100K	10K	240K	510K	3.6V	0mA

Battery options (non-rechargeable):

- Alkaline 2x1.5V or 3x1.5V (AAA, AA, B, C, D)
- Lithium Thionyl Chloride (Li-SOCl₂) 3.6V
- Lithium Manganese Dioxide (Li-MnO₂) 3.0V

Battery options (rechargeable):

- Li-Ion or Li-Pol
- 3xNiMH or 4xNiMH
- LiFePO₄

Formulas:

- $\text{R11: Maintenance current (Imin)}$
- $\text{Imin} = ((0.44/\text{R13}) + (0.44/\text{R11}) - (\text{VIN}/\text{R11})) \times 886$
- If Imin < 0, maintenance current is zero
- $\text{R13: Charge current} = 1950/\text{R13} = 195 \text{ mA}$
- $\text{R10, R12: Charge voltage} = 2.445 \times (1 + \text{R10}/\text{R12})$

[illegible]

Battery Monitoring

The diagram shows a circuit for battery monitoring. A green line represents the I035 IC. A red line represents the battery connection, starting from a terminal labeled 'VBAT' at the top. The red line goes through a resistor labeled 'R7 1M' to a green node. From this node, the red line goes through another resistor labeled 'R8 1M' to a green node. A capacitor labeled 'C7 100nF' is connected between the two green nodes. The red line then goes through a third resistor labeled 'R8 1M' to a green node. The red line ends at a battery symbol at the bottom. The green line starts at a green node, goes through a resistor labeled 'R7 1M' to a green node, then through a resistor labeled 'R8 1M' to a green node, and finally through a resistor labeled 'R8 1M' to a green node. The green line ends at a green node.

Mounting

M1 M2

M2.5 size

M3 M4

The diagram shows a simple LED circuit. A yellow LED, labeled D4, is connected to a 5V power source, labeled IO2. A resistor, labeled R9 with a value of 5K1, is placed in series with the LED to limit current. The circuit is completed by connecting the other end of the resistor to ground.

Notes

1. All capacitors are X7R 50V unless specified.
2. All resistors are 0.1W thick film 1% unless specified.

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